

System Analysis and Design

Eighth Edition

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Chapter 5

Data Modeling

Objectives

- Explain the rules and style guidelines for creating entity relationship diagrams (ERDs).
- Create an ERD.
- Describe the use of a data dictionary and metadata.
- Explain how to balance ERDs and data flow diagrams.
- Describe the process of normalization.

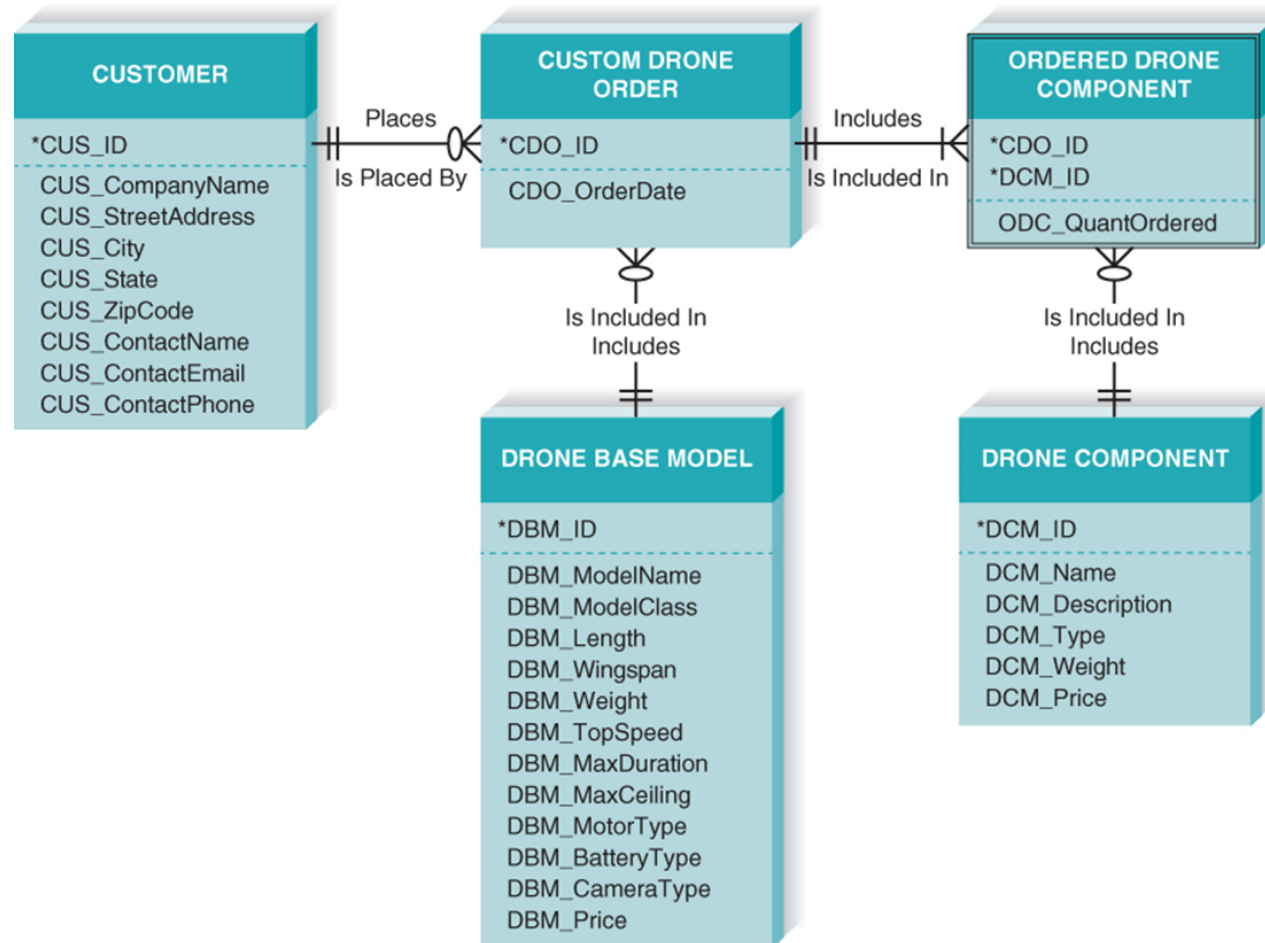
Introduction

- A ***data model*** is a formal way of representing the data that are used and created by a business system
 - Shows the people, places and things about which data is captured and the relationships among them
- ***Logical data model*** shows the organization of data without indicating how it is stored, created, or manipulated
- ***Physical data model*** shows how the data will actually be stored in databases or files

The Entity Relationship Diagram

- **Entity relationship diagram** (ERD) is a popular way to depict the data model
- On an ERD, similar kinds of information are listed together and placed inside boxes called **entities**
 - Lines are drawn between entities to represent relationships among the data
 - Special symbols are added to the diagram to communicate high-level business rules that need to be supported by the system
- The ERD implies no order

Reading an Entity Relationship Diagram



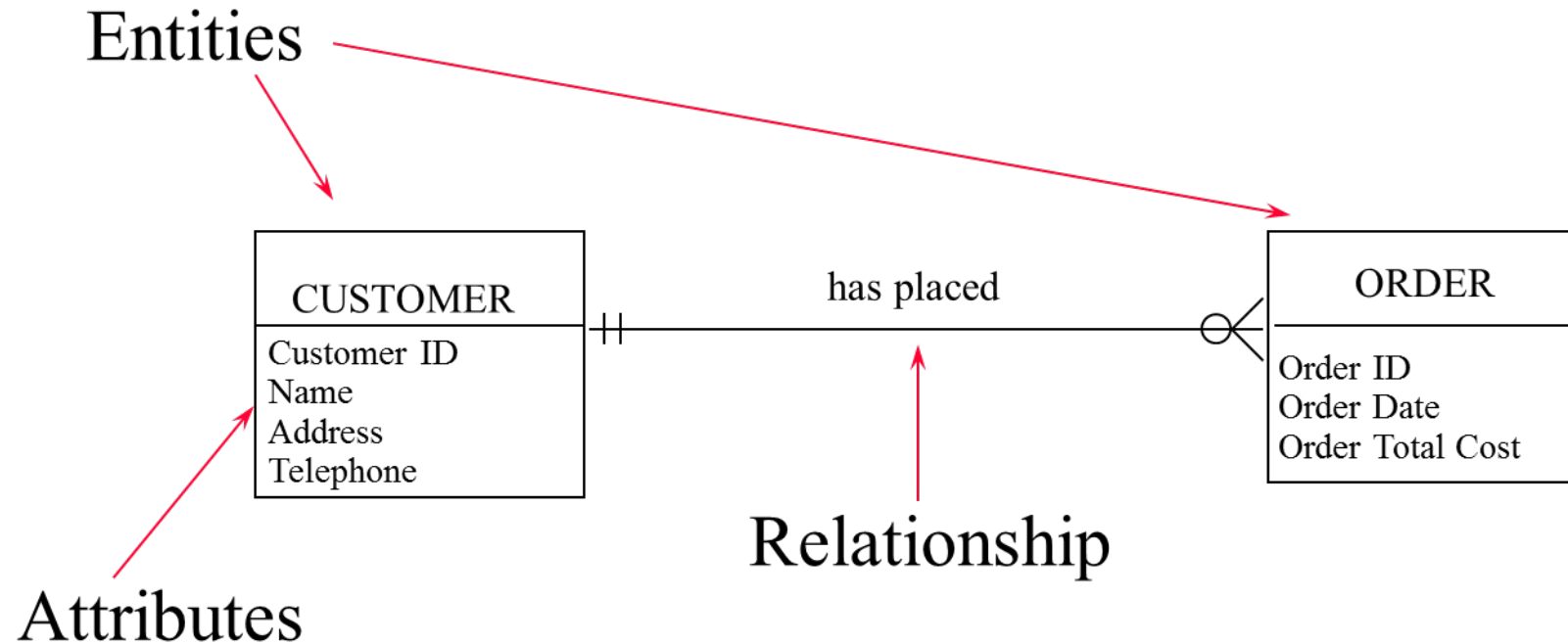
Using the ERD to Show Business Rules

- Business rules are constraints that are followed when the system is in operation
- ERD symbols can show when one instance of an entity must exist for an instance of another to exist
 - The customer instance must exist before custom drone order instances can be created for that customer
 - The drone component instance must exist before ordered drone components instances can be created for that drone component

Using the ERD to Show Business Rules Continued

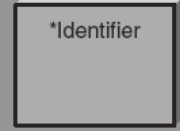
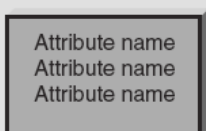
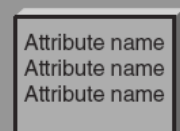
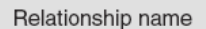
- ERD symbols can show when one instance of an entity can be related to only one or to many instances of another entity
 - One customer instance can create many custom drone order instances; each custom drone order instance is created by only one customer instance
 - A drone can be included on many chemical requests; a chemical request is for only one chemical
- ERD symbols show when the existence of an entity instance is optional for a related entity instance
 - A customer instance may exist without being included on any custom drone order instances

An ERD Example



Elements of an Entity Relationship Diagram

- Entity
- Attribute
- Relationships
- Cardinality
- Modality

	IDEF1X	Chen	Crow's Foot
An ENTITY ✓ is a person, place, or thing. ✓ has a singular name spelled in all capital letters. ✓ has an identifier. ✓ should contain more than one instance of data.	ENTITY NAME 	ENTITY NAME 	ENTITY NAME 
An ATTRIBUTE ✓ is a property of an entity. ✓ should be used by at least one business process. ✓ is broken down to its most useful level of detail.	ENTITY NAME 		ENTITY NAME 
A RELATIONSHIP ✓ shows the association between two entities. ✓ has a parent entity and a child entity. ✓ is described with a verb phrase. ✓ has cardinality (1 : 1, 1 : N, or M : N). ✓ has modality (null, not null). ✓ is dependent or independent.			

Entity

- A person, place, event, or thing about which data is collected
- Must be multiple occurrences to be an entity

Person	Place	Object	Event	Concept
STUDENT	DORM	BOOK	LIBRARY CHECK OUT	COURSE

CASE Entry for Entity

```
graph LR; CUSTOMER -- "Places / Is Placed By" --> ORDERED_DRONE_COMPONENT[ORDERED DRONE COMPONENT]; ORDERED_DRONE_COMPONENT -- "Is Included In / Includes" --> DRONE_BASE_MODEL[DRONE BASE MODEL];
```

Entry Description

Name:

Entry Type:

Description:

Attributes:

Name	Type	Length	Null
DBM_ID			
DBM_ModelName			
DBM_ModelClass			
DBM_Length			
DBM_Wingspan			
DBM_Weight			
DBM_TopSpeed			
DBM_MaxDuration			
DBM_MaxCeiling			
DBM_MotorType			
DBM_BatteryType			
DBM_CameraType			
DBM_Price			

Notes:

Attributes

- Information captured about an entity
- Only those used by the organization should be included in the model
- Attribute names are nouns
- Sometimes entity name is added at the beginning of the attribute name for clarity

CASE Entry for Attribute

Entry Description

Name:

DBM_ModelClass

Entry Type:

Data Element

Description:

This data element specifies the model classification for the base model drone. Model classes are based on the number of motors.

Values & Meanings:

Quad – drone has four motors
Hex – drone has six motors
Oct – drone has eight motors

Notes:

Default value: Quad.
The majority of our models today are four motor models.

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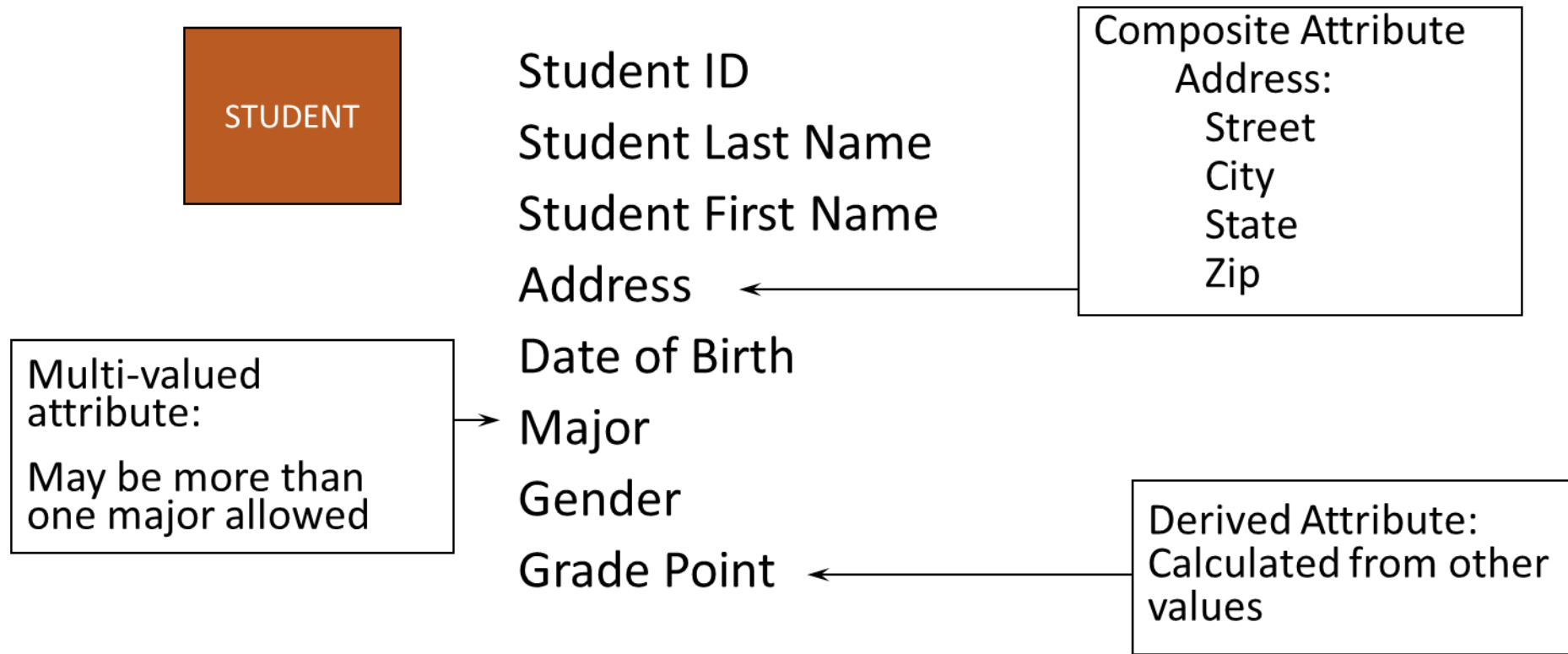
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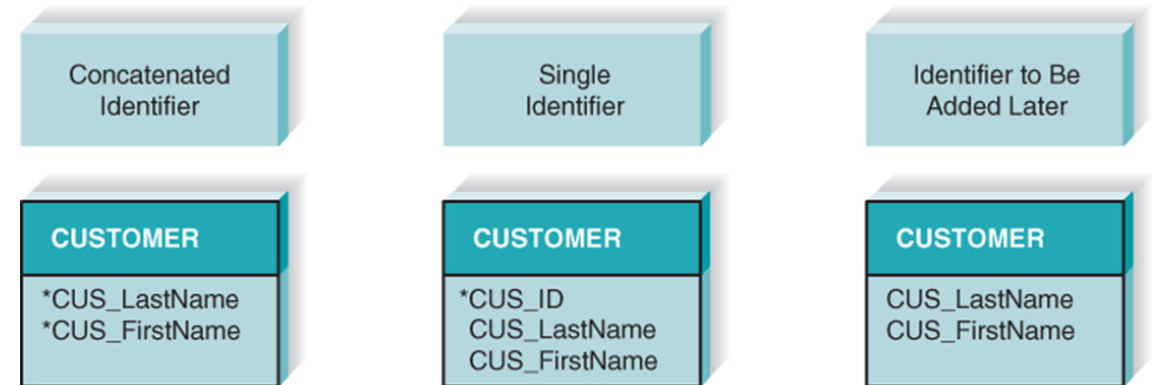
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Attributes – Understand Them!



Identifier Types

- One or more attributes can serve as the entity **identifier**, uniquely identifying each entity instance
- Concatenated identifier consists of several attributes
- An identifier may be “artificial,” such as creating an I D number
- Final decision on identifiers may postponed to the Design Phase



Relationships

- Associations between entities
- The first entity in the relationship is the ***parent*** entity; the second entity in the relationship is the ***child*** entity
- Relationships should have active verb names
- Relationships go in both directions

Cardinality

- Cardinality refers to the number of times instances in one entity can be related to instances in another entity
- One instance in an entity refers to one and only one instance in the related entity (**1:1**)
- One instance in an entity refers to one or more instances in the related entity (**1:N**)
- One or more instances in an entity refer to one or more instances in the related entity (**M:N**)

CASE Entry for Relationships

Entry Description

Name:

Places

Entry Type:

Relationship

Description:

Customer places zero or more Custom Drone Orders; a Custom Drone Order is placed by one and only one Customer

Attached Entities:

Customer	
Places	Min: 0 Max: Many
Custom Drone Order	
Is Placed By	Min: 1 Max: 1

Notes:

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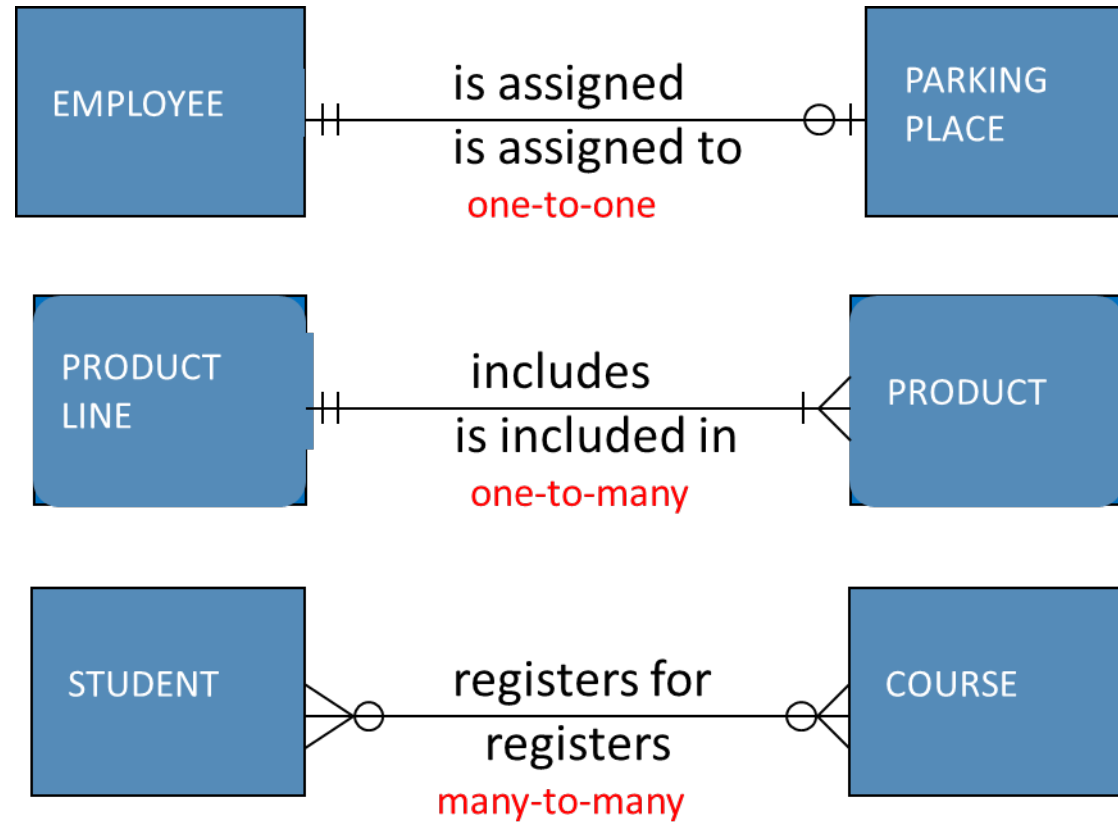
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Binary Relationships



Modality

- Refers to whether or not an instance of a child entity can exist without a related instance in the parent entity
- ***Not Null*** - an instance in the related entity must exist for an instance in another entity to be valid
- ***Null*** - no instance in the related entity is necessary for an instance in another entity to be valid

The Data Dictionary and Metadata

- Every CASE tool has a ***data dictionary***
 - Where the analyst goes to define or look up information about the entities, attributes, and relationships on the ERD
- The information you see in the data dictionary is called ***metadata***
 - Data about data
- Metadata are stored in the data dictionary so that they can be shared

```
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DBM_MaxDuration			
DBM_MaxCeiling			
DBM_MotorType			
DBM_BatteryType			
DBM_CameraType			
DBM_Price			

Notes:

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Types of Metadata Captured by the Data Dictionary

ERD Element	Kinds of Metadata	Example
Entity	Name	Item
	Definition	Represents any item carried in inventory in the supermarket
	Special notes	Includes produce, bakery, and deli items
	User contact	Nancy Keller (x6755) heads up the item coding department
	Analyst contact	John Michaels is the analyst assigned to this entity
Attribute	Name	Item_UPC
	Definition	The standard Universal Product Code for the item based on Global Trade Item Numbers developed by GS1
	Alias	Item Bar Code
	Sample values	036000291452; 034000126453
	Acceptable values	Any 12-digit set of numerals
	Format	12 digit, numerals only
	Type	Stored as alphanumeric values
	Special notes	Values with the first digit of 2 are assigned locally, representing items packed in the store, such as meat, bakery, produce, or deli items. See Nancy Keller for more information.
Relationship	Verb phrase	Included in
	Parent entity	Item
	Child entity	Sold item
	Definition	An item is included in zero or more sold items. A sold item includes one and only one item.
	Cardinality	1:N
	Modality	Null
	Special notes	

Creating an Entity Relationship Diagram

- Drawing the ERD is an iterative process of trial and revision
- ERDs can become quite complex
- Steps in building ERDs...
 1. Identify the entities
 2. Add appropriate attributes for each entity
 3. Draw the relationships that connect associated entities

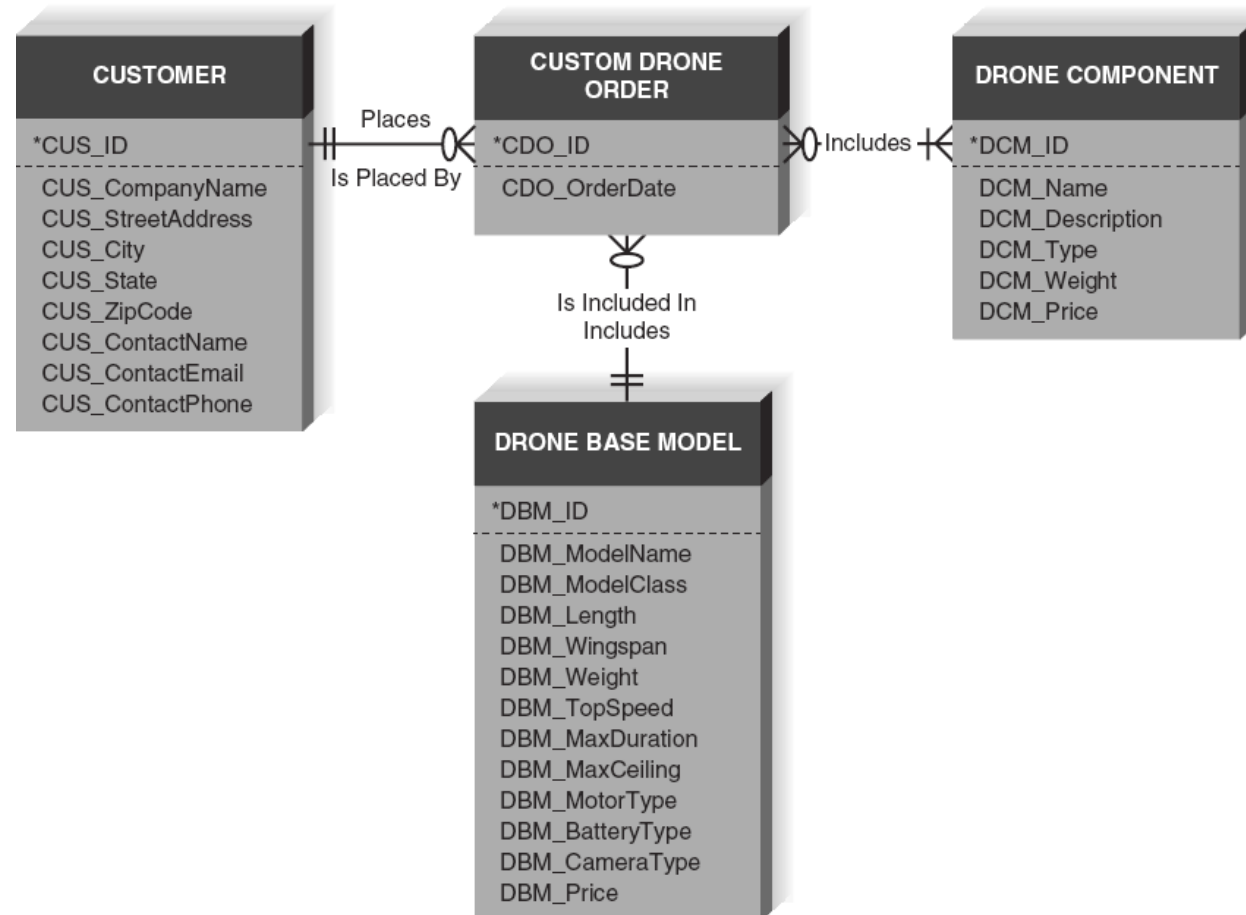
Identify the Entities

- Identify major categories of information
 - If available, check the process models for data stores, external entities, and data flows
 - Check the major inputs and outputs from the use cases
- Verify that there is more than one instance of the entity that occurs in the system

Add Attributes and Assign Identifiers

- Identify attributes of the entity that are relevant to the system under development
 - Check the process model repository entries for details on data flows and data stores
 - Check the data requirements of the requirements definition
 - Interview knowledgeable users
 - Perform document analysis on existing forms and reports
- Select the entity's candidate identifier (final decision may be postponed until Design phase)

Notice the Identifiers Selected for each Entity



Identify Relationships

- Start with an entity and identify all entities with which it shares relationships
- Describe the relationship with the appropriate verb phrase
- Determine the cardinality and modality by discussing the business rules with knowledgeable users

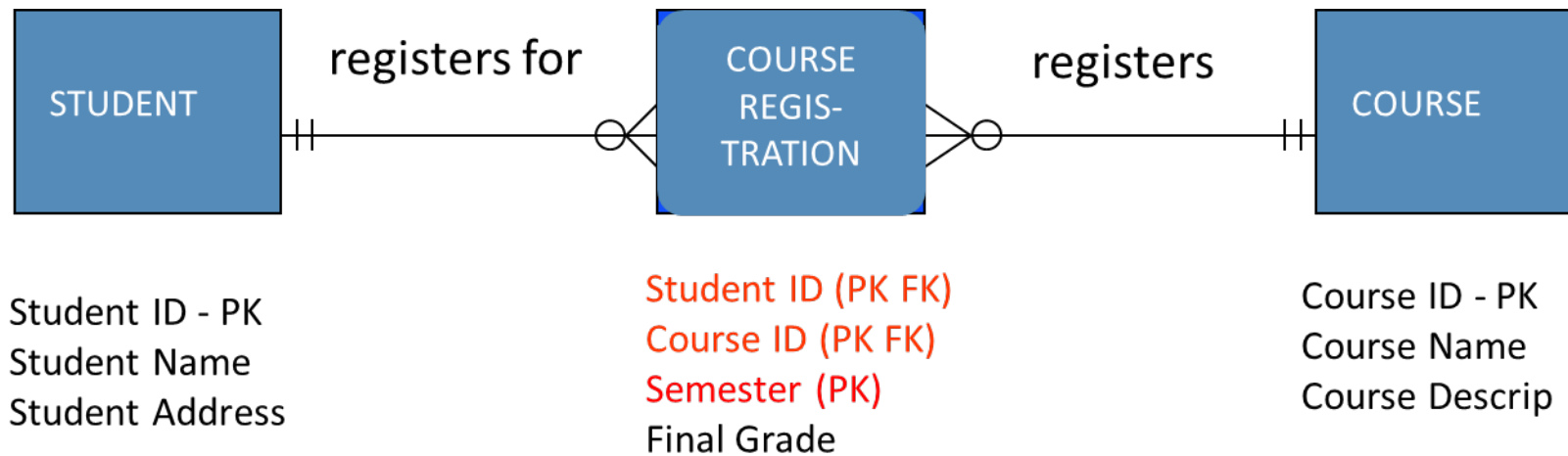
ERD Building Tips

- Data stores of the D F D generally correspond to entities
- Only include entities with more than one instance
- Do not include entities associated with implementation of the system (e.g., archive files of older data)
 - They will be added later

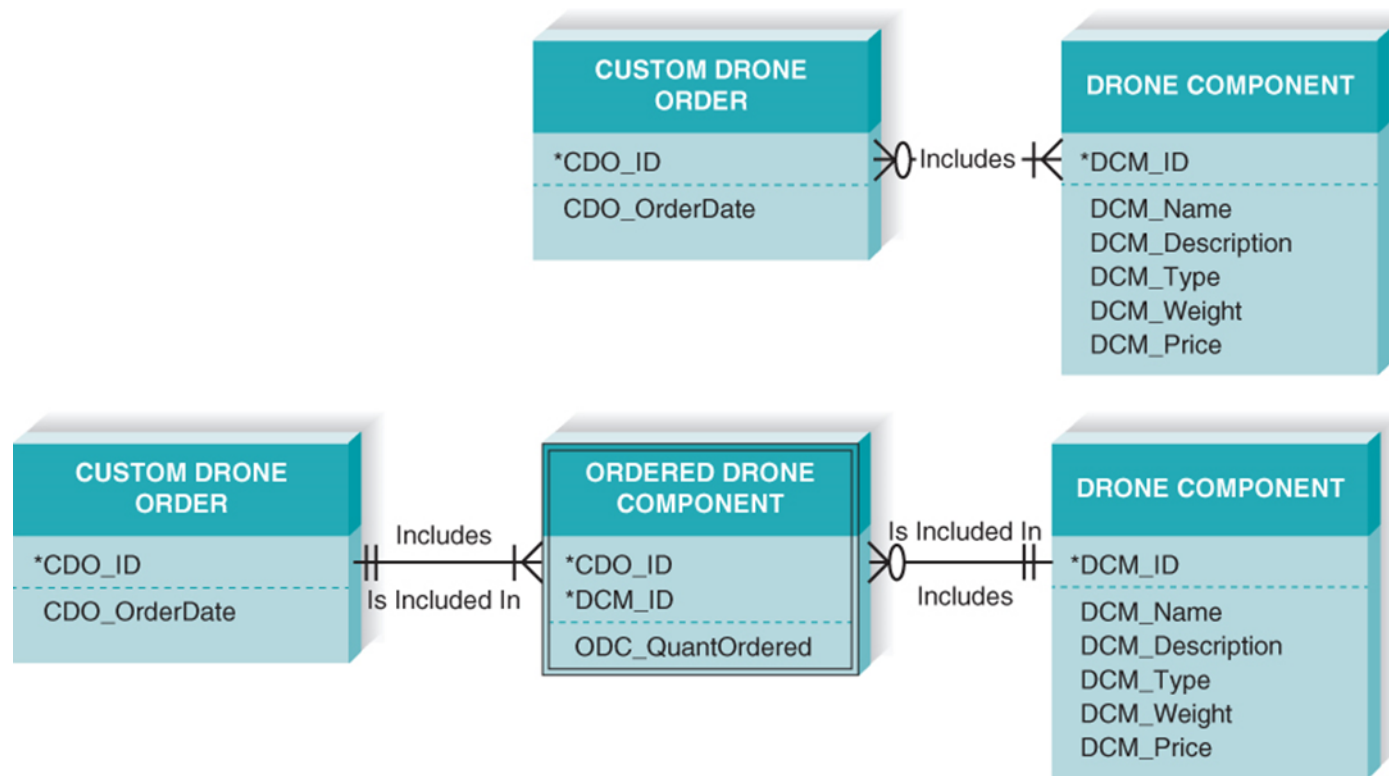
Advanced Syntax - Intersection Entities

- A new entity is created to store information about two entities sharing an M:N relationship
- Remove the M:N relationship between two entities and insert new entity between them
- Create two 1:N relationships: original entities are parents to the new child intersection entity
- Name the intersection entity
- Migrate parent entity primary keys to new entity as foreign keys (possibly also concatenated primary key)

Resolving M:N with an Intersection Entity



Resolving M:N Relationship



Other Syntax

- An ***independent entity*** is an entity that can exist without the help of another entity
- When a relationship includes an independent child entity, it is called a ***non-identifying relationship***
- There are situations when a child entity requires attributes from the parent entity to uniquely identify an instance
 - Called a ***dependent entity***
- When relationships have a dependent child entity, they are called ***identifying relationships***